

Section: SHS50x50x3 Height: 5.00 m

FAIL

Design Inputs

Height L = 5.00 m
Section: SHS50x50x3 (SHS)
Grade: Grade 350 $f_y = 350$ MPa
 $k_x = 1.00$ $L_{e_x} = 5.00$ m
 $k_y = 1.00$ $L_{e_y} = 5.00$ m
 $G = 300.0$ kN $Q = 200.0$ kN
 $N^* = 660.0$ kN
 $e_x = 0$ mm $e_y = 30$ mm
 $M^*_x = 19.80$ kN.m $M^*_y = 0.00$ kN.m

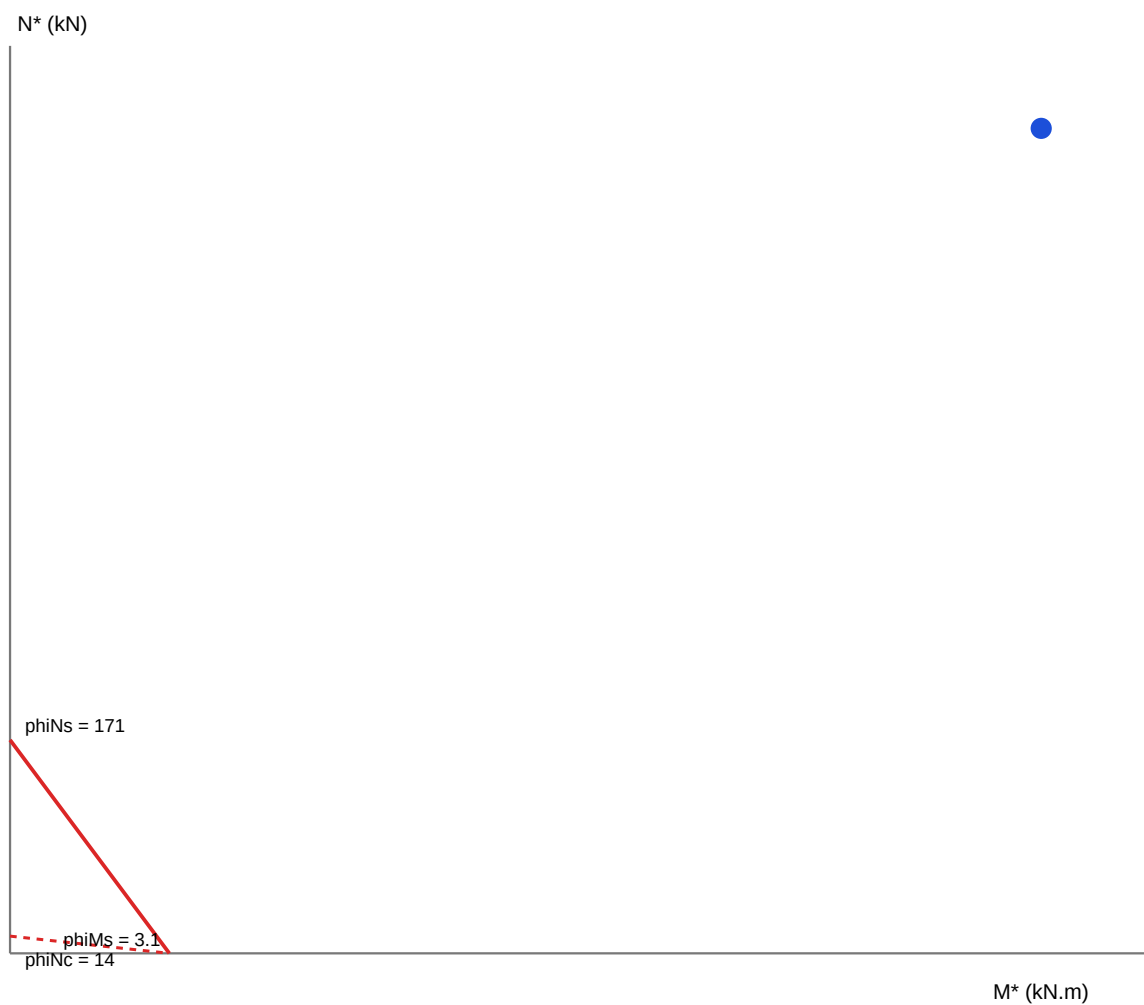
Section Properties

Designation SHS50x50x3
Mass 4.3 kg/m
 A_g 542.00 mm²
 I_x 2.06e+5 mm⁴
 I_y 2.06e+5 mm⁴
 r_x 19.5 mm
 r_y 19.5 mm

Capacity Check Summary

Check	Demand	Capacity	Util	Status
Axial - Section (ϕN_s)	660.0 kN	170.7 kN	386.6%	FAIL
Axial - Member x (ϕN_{c_x})	660.0 kN	13.8 kN	4792.4%	FAIL
Axial - Member y (ϕN_{c_y})	660.0 kN	13.8 kN	4792.4%	FAIL
Bending x - Section (ϕM_{sx})	19.80 kN.m	3.06 kN.m	647.3%	FAIL
Combined - Section (Cl. 8.3.3)	10.339	≤ 1.00	1033.9%	FAIL
Combined - Member in-plane (Cl. 8.4.2)	54.397	≤ 1.00	5439.7%	FAIL
Combined - Member out-of-plane (Cl. 8.4.4)	54.397	≤ 1.00	5439.7%	FAIL

N*-M*x Interaction Diagram (AS4100)



Solid line = section envelope (Cl. 8.3.3); dashed line = member envelope (Cl. 8.4).
Filled point = design action (M^* , N^*). Green = inside envelope; red = outside.

1. Steel grade and fy	[AS3678 / AS1163]
Grade Grade 350, fy = 350 MPa	
Hollow section - AS1163 flat fy (independent of thickness)	
2. Section classification	[AS4100 Cl. 5.2.2]
Section class: compact	
Closed section - compact at practical sizes/grades	
3. Effective section modulus	[AS4100 Cl. 5.2.3-5.2.5]
$Z_{e_x} = 9710.00 \text{ mm}^3$	
$Z_{e_y} = 9710.00 \text{ mm}^3$	
4. Bending section capacity	[AS4100 Cl. 5.2.1]
$\phi M_{sx} = 0.9 \times Z_{e_x} \times f_y / 1e6 = 3.06 \text{ kN.m}$	
$\phi M_{sy} = 0.9 \times Z_{e_y} \times f_y / 1e6 = 3.06 \text{ kN.m}$	
5. Lateral-torsional buckling	[AS4100 Cl. 5.6]
No LTB - closed section. $\phi M_{bx} = \phi M_{sx} = 3.06 \text{ kN.m}$	
6. Factored loads	[AS1170.1]
$N^* = \max(1.2G + 1.5Q, G) = \max(660.0, 300.0) = 660.0 \text{ kN}$	
$M^*_x = N^* \times e_y / 1000 = 660.0 \times 30 / 1000 = 19.80 \text{ kN.m}$	
$M^*_y = N^* \times e_x / 1000 = 660.0 \times 0 / 1000 = 0.00 \text{ kN.m}$	
7. Axial section capacity	[AS4100 Cl. 6.2.1-6.2.2]
$k_f = 1.00$ ($A_{eff} = A_g$)	
$\phi N_s = 0.9 \times k_f \times A_g \times f_y / 1000 = 0.9 \times 1.00 \times 542 \times 350 / 1000 = 170.7 \text{ kN}$	
8. Member capacity - x-axis	[AS4100 Cl. 6.3.3]
$\lambda_{n_x} = (L_{e_x} / r_{x}) \times \sqrt{k_f \times f_y / 250} = 303.5$	
$\alpha_{c_x} = 0.081$	
$\phi N_{c_x} = 0.9 \times \alpha_{c_x} \times k_f \times A_g \times f_y / 1000 = 13.8 \text{ kN}$	
9. Member capacity - y-axis	[AS4100 Cl. 6.3.3]
$\lambda_{n_y} = (L_{e_y} / r_{y}) \times \sqrt{k_f \times f_y / 250} = 303.5$	
$\alpha_{c_y} = 0.081$	
$\phi N_{c_y} = 0.9 \times \alpha_{c_y} \times k_f \times A_g \times f_y / 1000 = 13.8 \text{ kN}$	
$\phi N_c = \min(\phi N_{c_x}, \phi N_{c_y}) = 13.8 \text{ kN}$	
10. Combined - section	[AS4100 Cl. 8.3.3]
$N^* / \phi N_s + M^*_x / \phi M_{sx} + M^*_y / \phi M_{sy} = 10.339 \leq 1.0 \rightarrow \text{FAIL}$	
11. Combined - member in-plane	[AS4100 Cl. 8.4.2]
$N^* / \phi N_c + M^*_x / \phi M_{sx} = 54.397 \leq 1.0 \rightarrow \text{FAIL}$	
12. Combined - member out-of-plane	[AS4100 Cl. 8.4.4]
$N^* / \phi N_c + M^*_x / \phi M_{bx} = 54.397 \leq 1.0 \rightarrow \text{FAIL}$	
13. Overall result	[AS4100]
Overall: FAIL	